

DEWBERRY AGCM, AB, Canada

WMO: 713590

Lat: 53.6592N

Lon: 110.5875W

Elev: 2021

StdP: 13.65

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.4	-20.0	-31.9	1.0	-25.0	-26.0	1.4	-19.8	18.3	23.7	15.7	17.9	3.2	280	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	81.8	63.4	79.0	62.3	76.3	61.3	67.2	77.2	65.3	75.0	63.3	72.7	6.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	94.9	72.4	61.4	87.7	70.1	59.7	82.2	67.9	32.9	77.2	31.3	75.5	29.8	72.6	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.2	12.5	DB	-34.9	87.3	3.7	2.3	-37.5	89.0	-39.6	90.3	-41.7	91.6	-44.4	93.3	(4)
(5)			WB	-34.5	71.9	3.9	2.7	-37.3	73.9	-39.5	75.4	-41.7	77.0	-44.5	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	9.8	8.4	21.8	37.2	51.8	59.1	62.8	60.8	52.3	38.3	21.4	9.0		
	DBStd	22.98	15.39	15.00	13.44	9.62	8.15	5.34	4.33	5.20	8.39	8.66	12.64	14.19		
	HDD50	6339	1245	1164	875	394	79	5	0	2	72	372	860	1271		
	HDD65	10570	1710	1584	1340	835	413	188	94	147	386	828	1309	1736		
	CDD50	1304	0	0	0	9	134	279	398	336	140	8	0	0		
	CDD65	61	0	0	0	0	3	11	27	16	4	0	0	0		
	CDH74	1121	0	0	0	5	149	166	350	285	166	0	0	0		
	CDH80	177	0	0	0	0	21	21	51	41	43	0	0	0		
	WSAvg	5.3	5.6	5.4	5.7	6.4	5.9	5.4	4.6	4.2	4.7	5.6	5.2	5.0		
	PrecAvg	16.4	0.7	0.5	0.6	1.1	1.7	2.9	3.1	2.3	1.4	0.7	0.6	0.7		
(15) Precipitation	PrecMax	23.8	1.7	1.1	1.7	3.6	4.0	5.2	6.5	5.8	3.8	1.5	2.1	1.3		
	PrecMin	8.8	0.2	0.2	0.2	0.1	0.2	0.6	0.9	0.4	0.1	0.1	0.1	0.3		
	PrecStd	3.5	0.4	0.2	0.4	0.8	1.0	1.2	1.5	1.0	0.9	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	41.3	53.9	72.6	82.8	83.0	84.9	84.5	85.4	69.6	54.5	41.4		
		MCWB	37.4	36.2	42.6	50.3	56.9	62.0	69.0	64.6	61.9	54.1	43.2	35.3		
	2%	DB	39.0	37.2	46.9	65.0	78.4	78.6	80.9	80.3	78.9	61.2	46.2	37.0		
		MCWB	35.0	33.2	38.5	46.6	54.8	60.6	67.1	64.6	59.8	48.1	39.1	32.8		
	5%	DB	36.1	33.8	41.8	59.2	73.5	75.2	77.8	77.1	73.2	56.3	40.4	33.0		
		MCWB	32.9	30.9	36.4	44.4	53.4	59.2	64.9	62.8	57.4	46.1	35.8	29.6		
	10%	DB	32.3	30.8	38.2	53.4	69.3	72.0	75.1	74.2	67.9	51.7	36.6	28.5		
		MCWB	29.6	28.6	33.9	41.4	51.5	57.8	63.1	61.9	54.9	44.2	33.6	26.3		
	0.4%	WB	37.7	36.8	44.0	51.0	59.3	65.5	71.8	69.9	63.2	54.0	45.2	36.0		
		MCDB	42.2	40.6	52.6	72.0	75.1	77.1	82.6	79.6	80.5	66.9	52.8	40.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	33.7	39.2	47.5	56.7	63.1	68.4	66.7	60.8	50.2	39.9	33.0		
		MCDB	38.6	36.4	45.6	62.8	72.8	73.9	78.2	76.5	76.3	56.9	45.5	36.9		
	5%	WB	33.0	31.4	36.6	44.8	54.9	61.6	66.2	64.6	58.4	47.5	36.6	29.6		
		MCDB	36.0	33.8	41.4	58.0	69.9	71.2	75.7	74.4	71.3	55.1	39.8	32.5		
	10%	WB	29.5	28.2	34.3	42.3	53.1	59.6	64.4	62.6	55.8	44.5	33.6	26.3		
		MCDB	32.2	30.7	37.9	51.9	67.2	68.5	73.2	72.0	66.2	51.9	36.2	28.4		
	5% DB	MDBR	16.8	17.8	17.2	19.7	24.4	20.6	20.8	22.2	22.9	17.7	14.4	14.5		
		MCDBR	18.3	19.1	20.5	30.3	31.9	26.6	25.8	28.0	32.9	26.2	19.9	19.3		
		MCWBR	15.1	16.1	13.8	15.6	13.1	11.8	13.6	14.4	15.3	15.1	13.6	15.7		
		MCWBR	17.8	18.5	19.5	29.1	28.1	22.4	24.8	25.0	31.6	24.6	19.0	18.6		
(38) Clear-Sky Solar Irradiance	taub		0.243	0.291	0.282	0.328	0.348	0.367	0.363	0.373	0.310	0.275	0.250	0.216		
		taud	2.316	2.191	2.382	2.376	2.367	2.338	2.380	2.325	2.491	2.560	2.446	2.354		
	Ebn at Noon		239	254	290	287	285	279	279	268	277	267	239	233		
		Edh at Noon	17	28	29	34	36	37	35	35	26	19	16	14		
(44) All-Sky Solar Radiation	RadAvg		303	638	1118	1485	1842	1880	1935	1579	1111	637	304	223		
	RadStd		33	47	87	128	133	135	106	117	121	92	34	22		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page